Improving SLAM Reliability Through Multi-Sensor Time Synchronization on a Quadruped Robot

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Abstract

Reliable SLAM on legged robots depends on tight time synchronization between LiDAR, IMU, and camera sensors, since small timestamp errors compound into pose drift. On a Lite3 quadruped, FAST-LIVO2 SLAM failed within 20 seconds because its Livox Mid-360s LiDAR and IMU operate on separate, unsynchronized clocks, causing drift to grow by roughly one second per second until the algorithm stalled. Hardware sync was ruled out: PTP reaches the LiDAR but not the IMU oscillator, and the RealSense D435i's RGB and depth sensors sit on separate PCBs, making GPIO sync impossible for the camera. We instead implemented a software correction in `LIVMapper.cpp`: an Exponential Moving Average filter ($\alpha = 0.01$) continuously re-estimates the IMU-LiDAR offset, while the RealSense camera, whose driver already timestamps frames against the host clock via global-time mode, receives a separate, fixed offset from a YAML config rather than reusing the IMU's changing correction. Packet-drop traps were removed in favor of non-blocking warnings, and shared buffers were made thread-safe with `std::lock_guard`. The result was indefinite runtime instead of ∼20 seconds, zero IMU jump warnings after convergence, and complete rather than partial maps across office, basement, and outdoor tests, with 1.04 m of loop-closure drift over a 70.5 s loop and 1.74 cm (0.87%) room-width error. The correction reduces but does not eliminate timing uncertainty: sharp turns still cause camera-LiDAR misalignment, and drift accumulates unrecoverably in large environments. We recommend replacing the D435i with the RGB/depth-coplanar D415 for true hardware synchronization.

Problem

The Livox Mid-360s IMU and LiDAR operate on **separate hardware clocks**. Without correction:

- Drift grows ∼1 second per second
- FAST-LIVO2 SLAM fails after ∼20 seconds
- Maps are incomplete and inaccurate

Hardware sync not possible:

- PTP reaches LiDAR but **not** IMU oscillator
- D435i RGB/Depth on separate PCBs — GPIO sync impossible (Intel datasheet 337029-005)

Software Solution

Implemented an **Exponential Moving Average (EMA)** filter to continuously estimate and correct the IMU-LiDAR clock offset per IMU message:

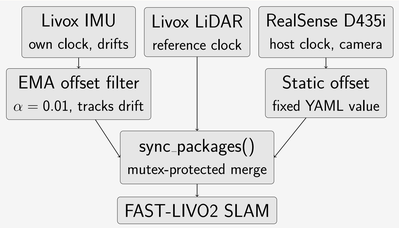
$$s_t = \alpha\, x_t + (1 - \alpha)\, s_{t-1}$$

where x_t is the measured lag, s_{t-1} is the previous estimate, and $\alpha = 0.01$ balances noise rejection against tracking speed.

Key changes to LIVMapper.cpp:

1. EMA filter in `imu_cbk`
2. Hard offset activated in `livox_pcl_cbk`
3. Camera clock decoupled from IMU offset
4. `std::lock_guard` throughout all callbacks
5. Removed IMU/image packet drop traps
6. Static reusable point cloud containers

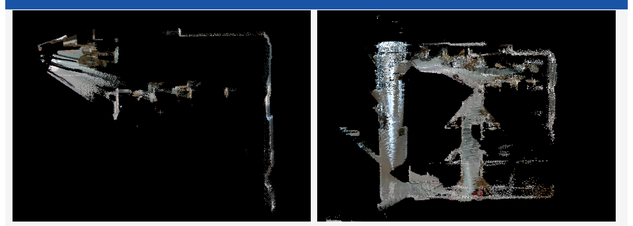
Sensor Clock Sync Pipeline



System Setup

- **Robot:** Lite3 quadruped (Deep Robotics)
- **LiDAR/IMU:** Livox Mid-360s
- **Camera:** Intel RealSense D435i
- **Compute:** NVIDIA Jetson AGX Orin
- **SLAM:** FAST-LIVO2 (ROS2 Humble)

Baseline vs Improved — Indoor Office



Run in a large office room with a central path and side corridor, starting and ending at the same location under normal indoor lighting. **Left:** the unsynchronized baseline dies after ∼10s once IMU-LiDAR lag exceeds tolerance, leaving a partial map. **Right:** with EMA sync active, the algorithm closes the loop and returns to start, though sharp turns still show minor camera-LiDAR misalignment.

Reconstruction Quality — Plant Test

Reconstructs a potted plant under normal lighting to compare completeness with and without soft-sync. **Top left:** baseline halts before finishing, leaving the pot base incomplete. **Top right:** with sync enabled, the pot and surrounding area are well captured. **Bottom:** real-life reference. Both struggle with fine leaf detail, a LiDAR resolution limit rather than a timing issue.

Key Results

Metric	Baseline	Improved
Runtime	∼20 s	Indefinite
IMU jump warnings	Hundreds	Zero
Map coverage	Partial	Complete
Loop closure drift	—	1.04 m / 70.5 s
Room width error	—	1.74 cm (0.87%)

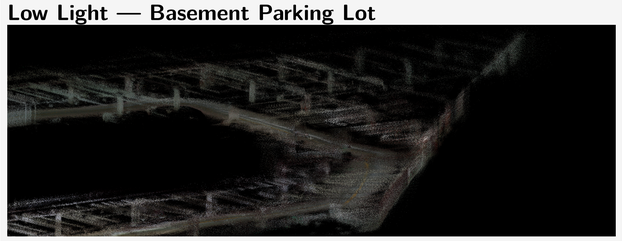
Baseline vs. EMA-corrected pipeline across core reliability metrics. The improved version runs indefinitely instead of failing at ∼20s, eliminates jump warnings after convergence, and produces complete maps with sub-2% error on independently verified measurements.

Reflective Floor — Marble

Conducted on glossy marble flooring to check whether specular window reflections would introduce false LiDAR geometry. Even with software sync active, the map correctly represents the reflection rather than treating it as an obstacle.

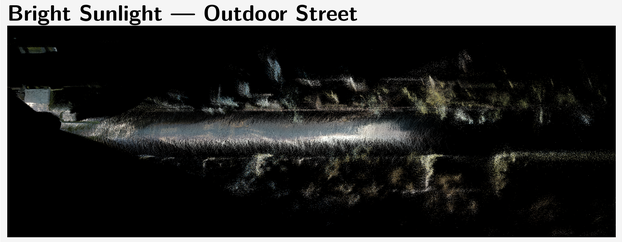
Environment Tests

Low Light — Basement Parking Lot



Conducted in an underground parking lot using slow movement. The map correctly shows the lot's contours and most pillars, though some are diffuse. Confirms LiDAR is unaffected by ambient light, while camera colour degrades in darkness.

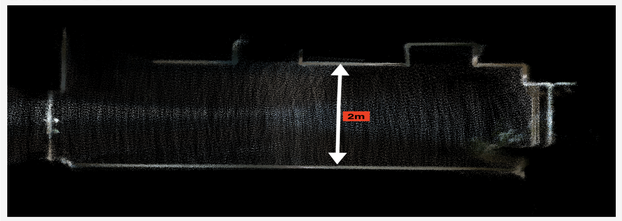
Bright Sunlight — Outdoor Street



Conducted outdoors in direct sunlight. Best colour fidelity and detail of any test, but high ambient temperature limited operation to ∼30s, producing a more fragmented map than the indoor tests.

Room Width Accuracy

Small room



A narrow elevator corridor was chosen for its known physical width, giving an independent ground-truth check. Width was measured in CloudCompare between two selected points. Measured corridor width in CloudCompare:

$$\begin{aligned} d_{map} &= \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2} \\ &= \sqrt{0.058 + 4.010 + 0.002} \\ &= 2.017 \text{ m} \end{aligned}$$

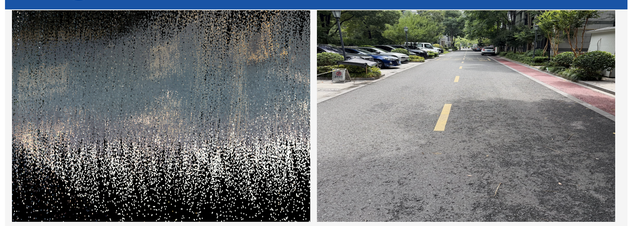
Real-world measurement: 2.000 m
Error: 1.74 **cm** (0.87%)

Loop Closure Drift

Measured over a closed trajectory in the indoor office test. Drift is the distance between the SLAM-estimated end position and the known start, measuring accumulated pose error. After a 70.5 second closed loop, the estimated end position vs physical start:

$$\begin{aligned} \mathbf{p}_{end} &= (-0.554,\ -0.875,\ -0.043) \text{ m} \\ d &= \|\mathbf{p}_{end} - \mathbf{p}_0\| \\ &= \sqrt{0.307 + 0.766 + 0.002} \\ &\approx \mathbf{1.04} \text{ m} \end{aligned}$$

Rough Ground — Street Pavement



Conducted on rough pavement to test accuracy on irregular ground. The map reproduces shadows and sunlight streaks correctly, but the camera angle and open environment limited reconstruction to a narrow strip rather than the full road width.

Limitations

- Soft-sync reduces but does not **eliminate** timing uncertainty
- Sharp turns cause camera-LiDAR frame misalignment
- Large environments: accumulated drift eventually halts mapping
- Robot overheats outdoors after ∼30 seconds

Future Work

- **Hardware IMU sync** via M12 connector Pin 8 (PPS signal)
- **Replace D435i** with Intel RealSense D415 for full RGB hardware sync
- Tune α parameter for different motion profiles
- Test in larger structured environments

Camera Replacement

	D435i	D415
RGB+Depth same PCB	×	✓
GPIO hardware sync	×	✓
External sync Pin 5	Depth only	RGB+Depth

The D415 was identified as a strong hardware-sync candidate: its RGB and depth sensors share one PCB, so both can be triggered from the same external sync pin, unlike the D435i's software-estimated global-time correction.

Project Repository

Scan for source code, branch `feature/wilhelm-imu-sync-ema`

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References

1. C. Zheng et al., "FAST-LIVO2: Fast, Direct LiDAR-Inertial-Visual Odometry," *IEEE Trans. Robotics*, vol. 41, pp. 326–346, 2025.
2. Intel Corp., "RealSense D400 Series Datasheet," Doc. 337029-005.
3. Livox Technology, "Mid-360 User Manual."